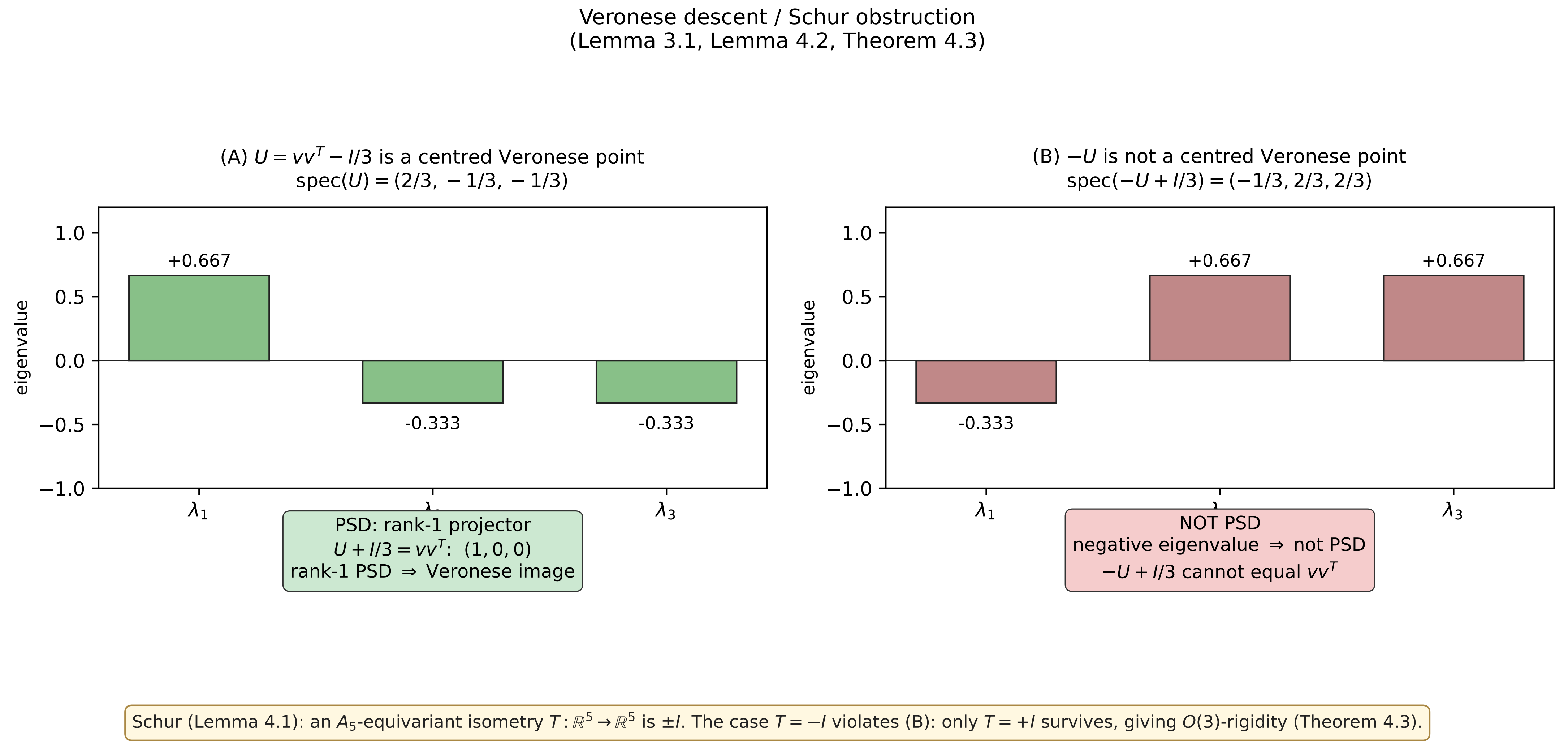
Veronese descent / Schur obstruction
(Lemma 3.1, Lemma 4.2, Theorem 4.3)
(A) $U = vv^T - I/3$ is a centred Veronese point
$\mathrm{spec}(U) = (2/3, -1/3, -1/3)$
eigenvalue
+0.667
-0.333
-0.333
λ_1
λ_3
PSD: rank-1 projector
$U + I/3 = vv^T$: $(1, 0, 0)$
rank-1 PSD $\Rightarrow$ Veronese image
(B) $-U$ is not a centred Veronese point
$\mathrm{spec}(-U + I/3) = (-1/3, 2/3, 2/3)$
eigenvalue
-0.333
+0.667
+0.667
λ_1
λ_3
NOT PSD
negative eigenvalue $\Rightarrow$ not PSD
$-U + I/3$ cannot equal vv^T
Schur (Lemma 4.1): an A_5-equivariant isometry $T: \mathbb{R}^5 \to \mathbb{R}^5$ is $\pm I$. The case $T = -I$ violates (B): only $T = +I$ survives, giving $O(3)$-rigidity (Theorem 4.3).